

ECE 3103 Spring 2025

Homework #1

(10 points)

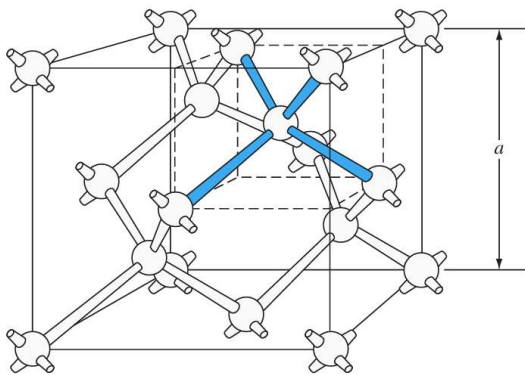
1. In our class notes we listed a few examples of semiconductors that crystalize in the diamond and zincblende structure. Pick three of them and find their bandgap energies (in eV) at room temperature. You can use Wikipedia, textbook, or any other 'trusted' sources. (1 point)

Semiconductor	Crystal Structure	Bandgap Energy (eV)
Galium arsenide (GaAs)	zincblende	1.42
Germanium (Ge)	diamond	0.66
Silicon (Si)	diamond	1.12

Source: Table B.4 - Neamen, Semiconductor Physics and Devices

2. Determine (a) the number of atoms per unit cell, (b) the number density of atoms (# of atoms per cm^3), and (c) the mass density (grams per cm^3), for silicon. The lattice constant of Si is $5.43\text{\AA}$. (3 points)

2a) number of atoms per unit cell

corners = $8 * 1/8$ piece of atom = 1 atomsides = $6 * 1/2$ piece of atom = 3 atomsinternal = $4 * 1$ atom = 4 atomstotal = $1 + 3 + 4$ **total = 8 atoms per unit cell**Problem 2b next page.

2b) number density of atoms

$$a = 5.43 \text{ \AA}$$

$$a = 5.43 * 10^{-8} \text{ cm}$$

$$\begin{aligned} \text{volume of unit cell} &= a^3 \\ &= (5.43 * 10^{-8} \text{ cm})^3 \\ &= 160.10 * 10^{-24} \text{ cm}^3 \\ \text{volume of unit cell} &= 1.601 * 10^{-22} \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{volume density of Si} &= \frac{\# \text{ atoms per unit cell}}{\text{volume of unit cell}} \\ &= \frac{8 \text{ atoms per unit cell}}{1.601 * 10^{-22} \text{ cm}^3} \\ \text{volume density of Si} &= 4.997 * 10^{22} \frac{\text{atoms}}{\text{cm}^3} \end{aligned}$$

2c) mass density

$$\text{molar mass of Si} = 28.085 \frac{\text{g}}{\text{mol}}$$

$$\begin{aligned} \text{mass of Si atom} &= \frac{\text{molar mass}}{1 \text{ mole}} \\ &= \frac{28.085 \frac{\text{g}}{\text{mol}}}{6.022 * 10^{23} \text{ atoms}} \\ \text{mass of Si atom} &= 4.66 * 10^{-23} \frac{\text{g}}{\text{atom}} \end{aligned}$$

$$\begin{aligned} \text{mass of unit cell} &= (\# \text{ atoms per unit cell}) * (\text{mass of Si atom}) \\ &= (8 \text{ atoms per unit cell}) * (4.66 * 10^{-23} \frac{\text{g}}{\text{atom}}) \end{aligned}$$

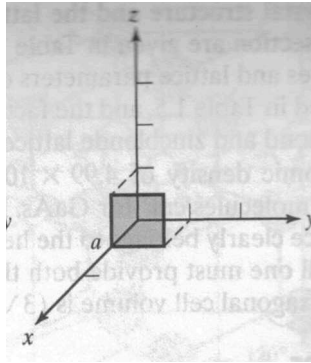
$$\text{mass of unit cell} = 3.728 * 10^{-22} \frac{\text{g}}{\text{unit cell}}$$

$$\begin{aligned} \text{mass density of Si} &= \frac{\text{mass per unit cell}}{\text{volume of unit cell}} \\ &= \frac{3.728 * 10^{-22} \frac{\text{g}}{\text{unit cell}}}{1.601 * 10^{-22} \text{ cm}^3} \end{aligned}$$

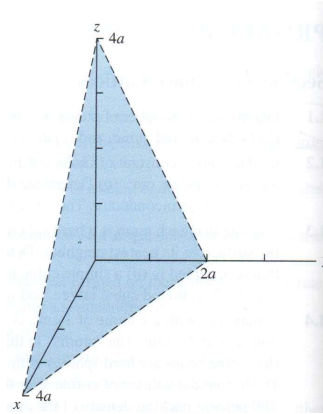
$$\text{mass density of Si} = 2.329 \frac{\text{g}}{\text{cm}^3}$$

Problem 3 next page.

3. For a simple cubic lattice, determine the Miller indices for the planes shown in the figures (a) and (b) below. (2 points)



(a)



(b)

RECIPE:

STEP 1) find intercepts of plane with the axes

STEP 2) write the reciprocals of the intercepts

STEP 3) multiply by lowest common denominator (LCD)

$$3a) \quad (1, \infty, \infty) \rightarrow (1/1, 1/\infty, 1/\infty) \rightarrow (1, 0, 0)$$

$$3b) \quad (4, 2, 4) \rightarrow (1/4, 1/2, 1/4) \rightarrow (1, 2, 1)$$

4. Determine the number of atoms per cm^3 in a diamond lattice with a lattice constant of $5.1 \text{ \AA}$. (1 point)

$$\# \text{ atoms per unit cell} = 8 \text{ atoms}$$

$$a = 5.1 \text{ \AA}$$

$$a = 5.1 \times 10^{-10} \text{ cm}$$

$$\begin{aligned} \text{volume of unit cell} &= a^3 \\ &= (5.1 \times 10^{-10} \text{ cm})^3 \\ &= 132.651 \times 10^{-30} \text{ cm}^3 \end{aligned}$$

$$\text{volume of unit cell} = 1.327 \times 10^{-22} \text{ cm}^3$$

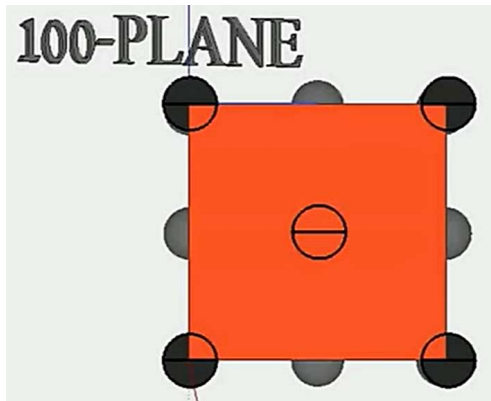
$$\begin{aligned} \text{volume density} &= \frac{\# \text{ atoms per unit cell}}{\text{volume of unit cell}} \\ &= \frac{8 \text{ atoms per unit cell}}{1.327 \times 10^{-22} \text{ cm}^3} \end{aligned}$$

$$\text{volume density} = 6.031 \frac{\text{atoms}}{\text{cm}^3}$$

Problem 5 next page.

5. The performance of some important semiconductor devices (incl. MOSFETs) depends upon the crystalline orientation of the wafer surface. Calculate the number of atoms per unit area (#atoms/cm²) for Si on each of the following planes: (100), (110), and (111). Use the fact that the lattice constant of Si is 5.43Å. (3 points)

Source of plane images: <https://www.youtube.com/watch?v=39H4QS1nv3E> (Lecture 15 January)



$$a = 5.43\text{\AA}$$

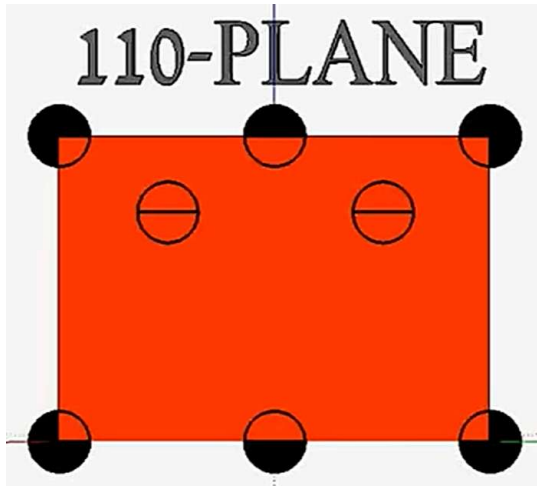
$$a = 5.43 * 10^{-8} \text{ cm}$$

$$\begin{aligned} \text{area of plane (100)} &= a^2 \\ &= (5.43 * 10^{-8} \text{ cm})^2 \\ \text{area of plane (100)} &= 2.948 * 10^{-15} \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{number of atoms per unit area (100)} &= \frac{\# \text{ atoms}}{\text{cm}^2} = \frac{\# \text{ atoms}}{\text{area of plane}} \\ &= \frac{2 \text{ atoms}}{2.948 * 10^{-15} \text{ cm}^2} \end{aligned}$$

$$\text{number of atoms per unit area (100)} = 6.783 * 10^{14} \frac{\text{atoms}}{\text{cm}^2}$$

Problem 5 continued next page



$$\sin(45^\circ) = \frac{a}{H} \rightarrow H = \frac{a}{\sin(45^\circ)}$$

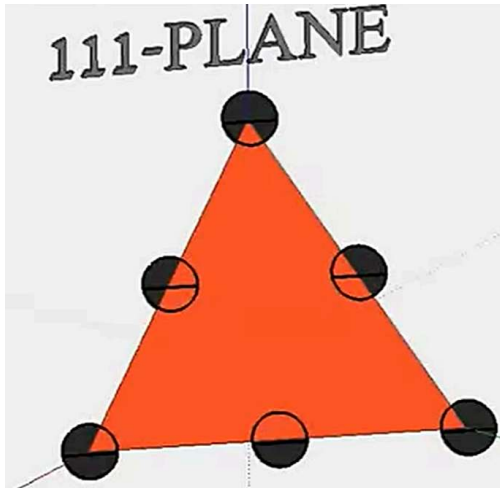
$$\begin{aligned} \text{area of plane (110)} &= a * H = \frac{a^2}{\sin(45^\circ)} \\ &= \frac{(5.43 \times 10^{-8} \text{ cm})^2}{1.414} \end{aligned}$$

$$\text{area of plane (110)} = 4.170 * 10^{-15} \text{ cm}^2$$

$$\begin{aligned} \text{number of atoms per unit area (110)} &= \frac{\# \text{ atoms}}{\text{cm}^2} = \frac{\# \text{ atoms}}{\text{area of plane}} \\ &= \frac{4 \text{ atoms}}{4.170 \times 10^{-15} \text{ cm}^2} \end{aligned}$$

$$\text{number of atoms per unit area (110)} = 9.593 * 10^{14} \frac{\text{atoms}}{\text{cm}^2}$$

Problem 5 continued next page



$$\begin{aligned} \text{area of plane (111)} &= \frac{3}{4}a^2 \\ &= \frac{\sqrt{3}}{4} (5.43 * 10^{-8} \text{ cm})^2 \end{aligned}$$

$$\text{area of plane (111)} = 1.277 * 10^{-15} \text{ cm}^2$$

$$\begin{aligned} \text{number of atoms per unit area (111)} &= \frac{\# \text{ atoms}}{\text{cm}^2} = \frac{\# \text{ atoms}}{\text{area of plane}} \\ &= \frac{2 \text{ atoms}}{1.277 * 10^{-15} \text{ cm}^2} \end{aligned}$$

$$\text{number of atoms per unit area (111)} = 1.567 * 10^{15} \frac{\text{atoms}}{\text{cm}^2}$$